

ARTICLE INFO

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ABSTRACT

Here goes the abstract

1. Introduction

In recent years there has been a global expansion of renewable power generation Shahbaz, Siddiqui, Siddiqui, Jiao and Kautish (2023); Hassan, Algburi, Sameen, Al-Musawi, Al-Jiboory, Salman, Ali and Jaszczur (2024), driven by de-carbonization efforts and declining costs in renewable technologies Jäger-Waldau (2025); Joshi, Rao and Banerjee (2025); Manservigi, Castorino, Spina, Venturini and Gonzalez-Salazar (2026). This rapid deployment of renewable generation has been mostly due to additions in solar (at both utility and residential scale), hydro and wind capacity Shahbaz et al. (2023); Hassan et al. (2024). Currently, annual renewable capacity additions outpace conventional capacity additions IRENA (2025), and this upward trend will likely continue in the forecoming decades Agency (2019); Raturi (2021).

However, the rapid expansion of renewable generation is leading to grid integration difficulties due to inherent variability and uncertainty in the output Li, Shi, Cao, Wang, Kuang, Tan and Wei (2015); Liu and He (2023). Spatiotemporal mismatches between power supply and demand may arise if capacity planning and grid infrastructure development do not keep pace Ming, Kun and Jun (2013); Bird, Lew, Milligan, Carlini, Estanqueiro, Flynn, Gomez-Lazaro, Holttinen, Menemenlis, Orths et al. (2016); O'Shaughnessy, Cruce and Xu (2020); Li, Wang, Guo, Xin, Zhang, Liu and Feng (2024). Oversupply of renewable generation — occurring during periods of low demand or high wind and solar resource availability — can result in curtailment of a fraction of the generated energy Denholm, O'Connell, Brinkman and Jorgenson (2015); Bird et al. (2016); O'Shaughnessy et al. (2020). This issue is further compounded by the increasing penetration of residential-scale distributed generation, which is generally excluded from centralized dispatch scheduling owing to its negligible individual capacity Li et al. (2015); Bhattarai, Maraseni and Apan (2022). As a result, these challenges place increasing demands on power system flexibility and energy storage capacity, both of which are essential for balancing supply and demand in high-penetration renewable

energy systems Denholm and Mai (2019); ?; Nejla, Saidani and Besbes (2025).

In particular, South America has also experienced accelerated penetration of renewable generation in recent years, with wind and solar photovoltaic capacity accounting for more than half of annual capacity additions International Energy Agency (2023). Although renewable penetration has been uneven across countries, several are already experiencing increasing curtailment rates von Papp, Moraga and Henríquez (2022); Silva, Renolphi, Vieira, Lourenço, Salles and Monaro (2025). Chile has recorded curtailment rates of up to 20% of total annual renewable generation, driven mainly by a geographical mismatch between solar generation — concentrated in the northern Atacama region — and demand centers located farther south von Papp et al. (2022); Silva et al. (2025). In Brazil, by contrast, the rapid expansion of distributed generation has reshaped the load profile, also resulting in high curtailment rates da Ponte, Rego and Júnior (2025); Silva, Vieira, Rego, Simone, Lourenço and Salles (2026). Despite these challenges, the region's abundant hydroelectric capacity Farfan and Breyer (2017); International Energy Agency (2023) represents an additional advantage for renewable energy integration, offering flexible storage and dispatch capabilities that can help mitigate variability and curtailment Wan and Parsons (1993); Hingray, Favre, Sauquet and Anquetin (2014).

Bolivia's current power mix is dominated by hydro and thermal generation—accounting for approximately 70% and 20% of installed capacity, respectively — with only a small share of non-hydro renewables Autoridad de Fiscalización de Electricidad y Tecnología Nuclear (AETN) (2026) —. Efforts are underway to further decarbonize the electricity sector: national development plans Viceministerio de Electricidad y Energías Renovables (2025) and Bolivia's Nationally Determined Contributions (NDC) Ministerio de Medio Ambiente y Agua and Autoridad Plurinacional de la Madre Tierra (2021) foresee substantial growth in centralized renewable capacity. The accelerated uptake of distributed solar PV observed in the region Chueca, Weiss, Celaya, Ravillard, Ortega, Tolmasquim and Hallack (2023) suggests that distributed generation growth represents a plausible complementary pathway.

Given this growing complexity in power systems, energy modeling has become a crucial tool for exploring

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supply and demand scenarios that support long-term policy decisions Huntington, Weyant and Sweeney (1982); Pfenninger, Hawkes and Keirstead (2014). However, few studies have modeled the impacts of renewable energy penetration on power systems in South America. In Bolivia, Fernandez-Vazquez et al. Fernandez Vazquez, Vansighen, Fernandez Fuentes and Quoilin (2022) developed a long-term optimization model of the Bolivian energy sector using OSeMOSYS, based on a trend-based projection of power demand, and analyzed policy-based scenarios to induce decarbonization of the power system. However, the specific impacts of renewable capacity expansion — particularly distributed generation — on transmission requirements, thermal fuel consumption, greenhouse gas (GHG) emissions, and energy curtailment remain unexplored. This study aims to analyze the impacts of exogenous renewable capacity additions — both centralized and distributed — on power transmission, fuel consumption in thermal power facilities, GHG emissions, and energy curtailment. An intraday hydroelectric dispatch strategy is also proposed to mitigate renewable energy curtailment. For this purpose, we constructed a regionalized model of the Bolivian energy sector using the LEAP and NEMO modeling tools.

The structure of the paper proceeds as follows: Section 2...

2. Methodology

2.1. Case study

Electricity demand currently represents approximately 11% of Bolivia's overall energy demand and has grown at an annual rate of approximately 5.7% in recent years ?? . This demand is predominantly met by natural gas (NG) and hydroelectric generation. By the end of 2024, the installed capacity of the National Interconnected System (NIS) totaled 3580 MW, of which 2573 MW corresponded to thermoelectric sources and 758 MW to hydroelectric sources, with the remaining 249 MW distributed across wind, solar, and biomass. Approximately 98% of thermoelectric capacity is NG-based, with the remainder corresponding to diesel engines; the NG-based fleet, in turn, is composed of 66% combined-cycle, 30% turbo-gas, and 4% turbo-vapor units.

NG-based generation accounts for most of Bolivia's capacity expansion over the last three decades ? . More recently, however, a steep decline in domestic NG production ? has motivated the development of national plans and policies aimed at decarbonizing the power sector ? Fernandez Vazquez et al. (2022). The capacity expansion plan assessed in this study, prepared by the Ministry of Energy, includes two ambitious scenarios of 100% renewable capacity additions up to 2037.

Bolivia's NIS has no current linkages with the national grids of bordering countries and can therefore be considered an isolated system; it currently covers eight out of nine departments. Given this configuration, Bolivia's energy system was represented in this study as a multi-nodal system, consisting of five nodes or regions obtained by grouping the administrative divisions as follows:

Table 1
Power lines

Line	Nodes	Capacity [MW]
L1	Centro-Norte	390
L2	Centro-Sur	260
L3	Centro-Oriente	165
L4	Oriente-Norte	35
L5	Centro-Oriente	500

- Pando
- Norte (La Paz + Beni)
- Centro (Cochabamba + Oruro)
- Oriente (Santa Cruz)
- Sur (Tarija, Potosí, and Chuquisaca)

This regionalization was based on prior energy modeling studies conducted in Bolivia that proposed this specific departmental aggregation (see ?). Accordingly, this structure was replicated in the present study to facilitate comparison with independently conducted modeling exercises.

Figure 1 shows the location of the nodes and the representation of the NIS, for which two power lines connecting Centro and Oriente, one line connecting Centro and Sur, one line connecting Centro and Norte, and one connecting Oriente and Norte were considered. Pando Region is not currently interconnected with the other regions. AGREGAR LOSSES The main features of the power lines are detailed in Table 1.

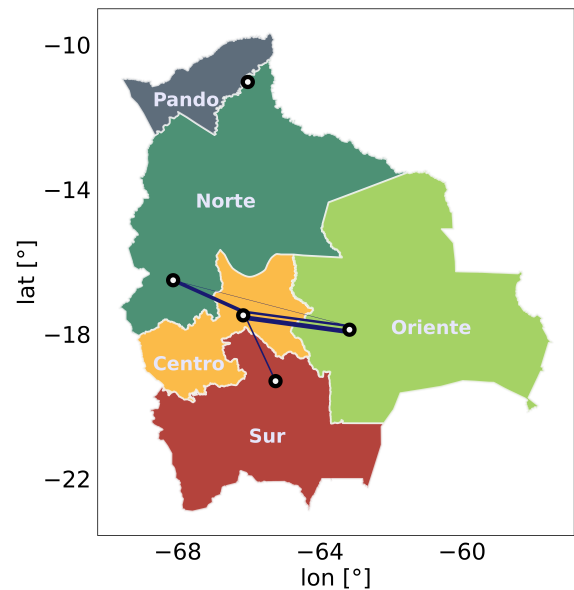


Figure 1: Regional nodes included in the model and power lines

2.2. Modelling approach

In this study, the Low Emissions Analysis Platform (LEAP) and the Next Energy Modeling System for Optimization (NEMO) were used to model Bolivia's energy sector. LEAP is a modeling tool developed by the Stockholm Environment Institute ². It has been widely used by researchers and organizations to project future energy demand and supply and greenhouse gas (GHG) emissions, and to assess energy policies and scenarios ^{22,23}. It supports the implementation of multiple modeling approaches within a single model, including bottom-up ² and top-down methodologies ², and it allows the formulation and comparison of different energy scenarios. NEMO is an open-source tool for energy system optimization ². Combined with LEAP, it enables additional modeling capabilities, such as least-cost optimization of energy supply and demand, multi-regional trade simulation, nodal network configuration, and energy storage modeling. In this study, we implemented a bottom-up approach and enabled NEMO's optimization module, together with the Cbc solver, for non-renewable capacity expansion, nodal simulation, and electricity dispatch.

The structure of both energy demand and supply — including regional features — was replicated in LEAP using information from national energy balances and inventories ²⁴ Autoridad de Fiscalización de Electricidad y Tecnología Nuclear (AETN) (2026), which provide annual energy demand volumes disaggregated by sector (industry, residential, transport, etc.) and by energy source (electricity, gasoline, diesel, etc.), as well as an updated generation fleet inventory. The base year of the simulation was set to 2023, and the simulation covers the 2023–2050 period.

2.3. Demand

As mentioned before, energy demand was modeled for 5 geographical regions (Pando, Norte, Centro Oriente, Sur) and for 7 sectors: residential, transport, industry, commercial + public, agriculture + fishery + mining, and construction. For the residential sector, the number of households represents the unit of consumption. We used population projections from the national bureau of statistics (INE in spanish ²⁵) to calculate the projected number of households. Households were then disaggregated into urban and rural subsectors, and rural households were further divided into electrified and non-electrified. Final energy uses — such as cooking, water heating, food conservation, and lighting — were then characterized in terms of coverage, efficiency, fuel share, and energy intensity (DE DONDE SALIO ESTO, CITAR). Miscellaneous thermal and electrical uses were also characterized. All electrical appliances were associated with a load curve (EN QUE TRABAJO PREVIO SE CARACTERIZO ESTO).

The transport sector was sub-divided into: road (light-duty, medium-duty, and heavy-duty), rail, air, waterway, and cable car. The vehicle fleet was regionalized according to inventories provided by INE.

The industrial sector was disaggregated into large-scale and small-scale units. For small-scale units, previous sectoral surveys (CITA) allowed further disaggregation into seven categories: food, beverages and tobacco; non-metallic minerals; paper, wood and furniture; machinery and metals; textiles; clothing and leather; and chemicals, plastics and pharmaceuticals. Large-scale units were classified into seven categories according to a previous sectoral survey from INE: food and beverage; cement; bricks; metals; paper, cardboard and wood; chemicals and petrochemicals; and construction materials. For this disaggregation level, we included the consumption share of different fuel types for the years 2021–2023. The fuels considered are electricity, diesel, gasoline, jet fuel, biomass, natural gas (NG), and liquefied petroleum gas (LPG).

Only one energy demand growth scenario was considered in this modeling exercise. For the transport; industry; commercial and public; agriculture, fishery and mining; and construction sectors, gross domestic product (GDP) was taken as the driver of energy demand. Sub-national reports from INE allowed the regional and sectoral disaggregation of GDP. An annual growth rate of 2.8%, based on historical trends, was considered for the simulation period. For the residential sector, the number of households was taken as the driver of energy demand. INE projects an overall population growth of 20%, reaching a total of 13.4 million inhabitants by 2050. Data from the last census allowed the regional and urban/rural disaggregation of population. For each region, both total population and number of households were extrapolated from the base year onwards, assuming that the regional proportion remains constant — an assumption that held for the 2012–2024 period. In order to project the regional urban/rural share of households, the ratio of urban-to-rural growth rates was computed. Then, an objective function was included in the projection so that the overall projected national number of urban households matched CEPAL's projection.

An analysis of the current energy demand structure shows that electricity represents approximately 11.8% of the overall energy demand. Electricity demand within the NIS is concentrated in the Oriente region (39%), followed by Centro, Norte (22% each), and Sur (17%) (Figure 1, panel a). The most electricity-consuming sector is residential (37%), followed by industry (24%); commercial and public (23%); agriculture, fishery and mining (11%); construction (3%); and transport (0.4%).

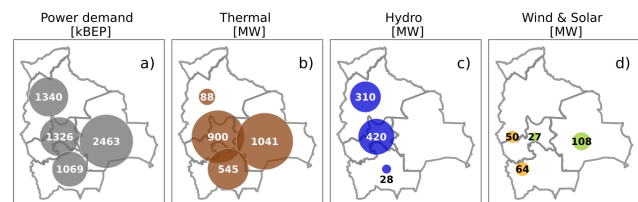


Figure 2: Regional nodes included in the model and power lines

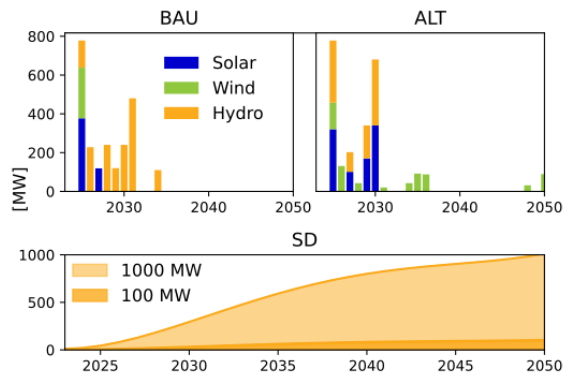


Figure 3: Regional nodes included in the model and power lines

Table 2

CRediT authorship contribution statement

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